

Prior-Data Fitted Networks

[s26] Theoretical Foundations of Amortization and Meta-, and In-Context Learning

Egor Kashkarov

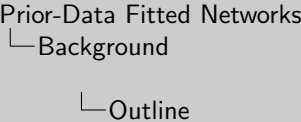
July 3, 2026

- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

Main content area for the presentation slide.

- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

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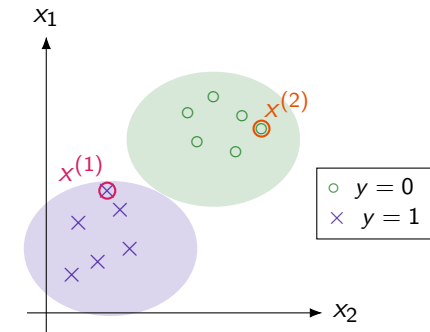


Outline

- Background
- TabPFN
- Statistical Foundations of PFNs
- TabPFNV2 & TabPFNV2.5
- TabICL
- Recent SOTA Models
- Connections to Other Topics
- Conclusion

Background

Statistical Model



$$\text{e.g. } D_{12} = \left\{ (x^{(1)}; 1), (x^{(2)}; 0), \dots \right\}$$

X – features; Y – classification label

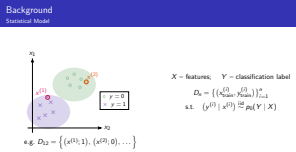
$$D_n = \left\{ (x_{\text{train}}^{(i)}, y_{\text{train}}^{(i)}) \right\}_{i=1}^n$$
$$\text{s.t. } (y^{(i)} | x^{(i)}) \stackrel{\text{iid}}{\sim} p_0(Y | X)$$

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Prior-Data Fitted Networks

└ Background

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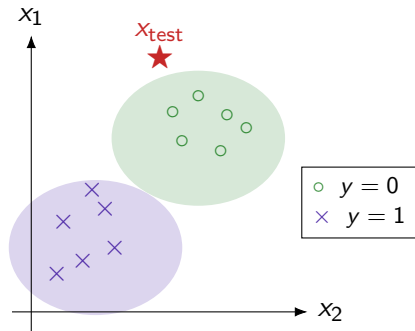


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 $D_n = \left\{ (x_{\text{train}}^{(i)}, y_{\text{train}}^{(i)}) \right\}_{i=1}^n$
s.t. $(y^{(i)} | x^{(i)}) \stackrel{\text{iid}}{\sim} p_0(Y | X)$

e.g. $D_{12} = \left\{ (x^{(1)}; 1), (x^{(2)}; 0), \dots \right\}$

Background

Supervised Learning Framework



At inference we want to estimate $p_0(y \mid x)$:

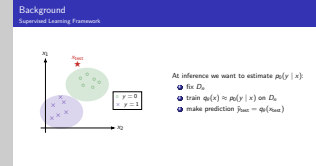
- 1 fix D_n
- 2 train $q_\theta(x) \approx p_0(y \mid x)$ on D_n
- 3 make prediction $\hat{y}_{\text{test}} = q_\theta(x_{\text{test}})$

2026-07-03

Prior-Data Fitted Networks

└ Background

└ Background



Want to estimate

$$\pi(y_{\text{test}} \mid x_{\text{test}}, D_n) \stackrel{\text{def}}{=} \mathbb{P}(Y = y_{\text{test}} \mid X = x_{\text{test}}; D_n)$$

With

$$q_{\theta}(x_{\text{test}}, D_n) \approx \pi(x_{\text{test}}, D_n) \quad \text{and} \quad \hat{y}_{\text{test}} = q_{\theta}(x_{\text{test}}, D_n)$$

π is called the **Posterior Predictive Distribution**

2026-07-03

Prior-Data Fitted Networks

- └ Background
- └ Background

Background

Bayesian Framework

Want to estimate

$\pi(y_{\text{test}} \mid x_{\text{test}}, D_n) \stackrel{\text{def}}{=} \mathbb{P}(Y = y_{\text{test}} \mid X = x_{\text{test}}; D_n)$

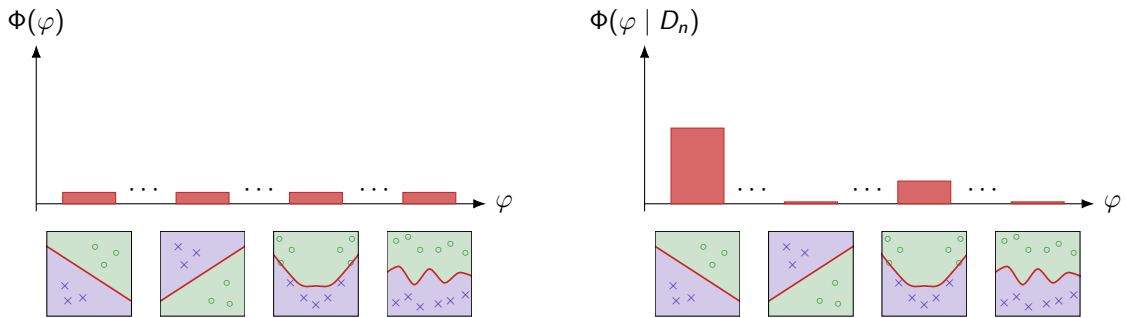
With

$q_{\theta}(x_{\text{test}}, D_n) \approx \pi(x_{\text{test}}, D_n) \quad \text{and} \quad \hat{y}_{\text{test}} = q_{\theta}(x_{\text{test}}, D_n)$

π is called the **Posterior Predictive Distribution**

Prior Φ

Define a hypothesis space Φ of data generation mechanisms where each hypothesis $\varphi \in \Phi$: $(y \mid x) \sim \varphi$.

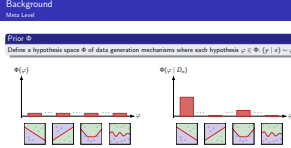


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Prior-Data Fitted Networks

└ Background

└ Background



$$\pi(y_{\text{test}} \mid x_{\text{test}}, D_n) = \int \varphi(y \mid x) d\Phi(\varphi \mid x, D_n) \tag{1}$$

2026-07-03

Prior-Data Fitted Networks

- └ Background
- └ Background

Background

Posterior Predictive Distribution

$$\pi(y_{\text{test}} \mid x_{\text{test}}, D_n) = \int \varphi(y \mid x) d\Phi(\varphi \mid x, D_n) \tag{1}$$

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$$\begin{aligned} &= \mathbb{P}(\varphi = \varphi_1 = \text{[diagonal split box]} \mid x_{\text{test}}, D_n) \cdot \varphi_1(y_{\text{test}} \mid x_{\text{test}}) \\ &+ \mathbb{P}(\varphi = \varphi_2 = \text{[diagonal split box]} \mid x_{\text{test}}, D_n) \cdot \varphi_2(y_{\text{test}} \mid x_{\text{test}}) + \dots \end{aligned}$$

2026-07-03

Prior-Data Fitted Networks

- └ Background
- └ Background

Background

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$$\pi(y_{\text{test}} \mid \textcolor{red}{x}_{\text{test}}, D_n) = \int \varphi(y \mid x) d\Phi(\varphi \mid x, D_n) \tag{1}$$

$$\begin{aligned} &= \mathbb{P}(\varphi = \varphi_1 = \text{img} \mid \textcolor{red}{x}_{\text{test}}, D_n) \cdot \varphi_1(y_{\text{test}} \mid x_{\text{test}}) \\ &+ \mathbb{P}(\varphi = \varphi_2 = \text{img} \mid \textcolor{red}{x}_{\text{test}}, D_n) \cdot \varphi_2(y_{\text{test}} \mid x_{\text{test}}) + \dots \end{aligned}$$

Assumption

$\mathbb{P}(\varphi \mid x, D) = \mathbb{P}(\varphi \mid D)$ (the hypothesis does not depend on x_{test} itself)

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2026-07-03

Prior-Data Fitted Networks

└ Background

└ Background

Background

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Background

Posterior Predictive Distribution (2)

Dropping x_{test} from the conditioning on φ :

$$\pi(y_{\text{test}} \mid x_{\text{test}}, D_n) = \int \varphi(y \mid x) d\Phi(\varphi \mid D_n) = \int p(y_{\text{test}} \mid x_{\text{test}}, \varphi) \cdot \underbrace{p(\varphi \mid D_n)}_{\text{not available}} d\varphi$$

2026-07-03

- Prior-Data Fitted Networks
 - Background
 - Background

Background

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2026-07-03

Prior-Data Fitted Networks

└ Background

└ Background

Background

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Background

Prior-Data Fitted Networks (PFNs)

Prior-Data Fitted Network is a numerical approximation of the entire *family* of PPDs $\{\pi(\cdot \mid \cdot, D_n)\}_n$.

Theorem 2.2. [1]

Let $\mathcal{Q}$ be the set of *all* conditional probability functions. Then the PPD satisfies

$$\pi = \arg \max_{q \in \mathcal{Q}} \mathbb{E}_{\varphi \sim \Phi} \left[\log q(Y \mid X, D_n) \right].$$

⇓ restrict to a parametric model q_θ

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{N \sim \Phi_N, \varphi \sim \Phi} \left[\log q_\theta(Y \mid X, D_N) \right]$$

⇓ approximate in practice by Monte-Carlo

$$\hat{\theta} = \arg \max_{\theta} \sum_{j=1}^m \log q_\theta(Y_j \mid X_j, D^{(j)})$$

2026-07-03

Prior-Data Fitted Networks

└ Background

└ Background

The expectation in Theorem 2.2 follows the hierarchical data-generating mechanism: first draw the true model (data-generating process) $\varphi \sim \Phi$; then generate the training sample D_n and the test pair (Y, X) from that selected φ .

Background

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- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

2026-07-03

Prior-Data Fitted Networks
└ TabPFN
└ Outline

Outline

- Background
- TabPFN
- Statistical Foundations of PFNs
- TabPFNV2 & TabPFNV2.5
- TabICL
- Recent SOTA Models
- Connections to Other Topics
- Conclusion

Conclusions

...

Limitations

...

2026-07-03

Prior-Data Fitted Networks

└ TabPFN

└ TabPFN

TabPFN

Discussion

Conclusions

...

Limitations

...

- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

2026-07-03

Prior-Data Fitted Networks
└ Statistical Foundations of PFNs
└ Outline

Outline

- Background
- TabPFN
- Statistical Foundations of PFNs
- TabPFNV2 & TabPFNV2.5
- TabICL
- Recent SOTA Models
- Connections to Other Topics
- Conclusion

Statistical Foundations of PFNs

Studied Questions

2026-07-03

Prior-Data Fitted Networks

Statistical Foundations of PFNs

—Statistical Foundations of PFNs

Statistical Foundations of PFI

Statistical Foundations of PFNs

When PPDs can Learn

Statistical Foundations of PFNs

2026-07-03

Prior-Data Fitted Networks

Statistical Foundations of PFNs

—Statistical Foundations of PFNs

Statistical Foundations of PFNs

PFN Approximation of the PPD

Statistical Foundations of PFNs

2026-07-03

Prior-Data Fitted Networks

—Statistical Foundations of PFNs

Statistical Foundations of PFNs

Statistical Foundations of PFNs

Bias-Variance Decomposition of PFNs

Statistical Foundations of PFNs

2026-07-03

Prior-Data Fitted Networks

Statistical Foundations of PFNs

— Statistical Foundations of PFNs

Statistical Foundations of PFNs

Bias and the Need for Locality

Statistical Foundations of PFNs

2026-07-03

Prior-Data Fitted Networks

Statistical Foundations of PFNs

— Statistical Foundations of PFNs

2026-07-03

Prior-Data Fitted Networks

Statistical Foundations of PFNs

—Statistical Foundations of PFNs

Statistical Foundations of PFNs

Localized PFNs

Statistical Foundations of PFNs

2026-07-03

Prior-Data Fitted Networks

- Statistical Foundations of PFNs

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2026-07-03

- Statistical Foundations of PFNs

—Statistical Foundations of PFNs

- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

2026-07-03

Prior-Data Fitted Networks
└ TabPFNV2 & TabPFNV2.5
└ Outline

Outline

- Background
- TabPFN
- Statistical Foundations of PFNs
- TabPFNV2 & TabPFNV2.5**
- TabICL
- Recent SOTA Models
- Connections to Other Topics
- Conclusion

Background

Definition

Insert key definition here.

Key insight

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2026-07-03

- Prior-Data Fitted Networks
 - TabPFNv2 & TabPFNv2.5
 - Background

Background
Definition
Insert key definition here.
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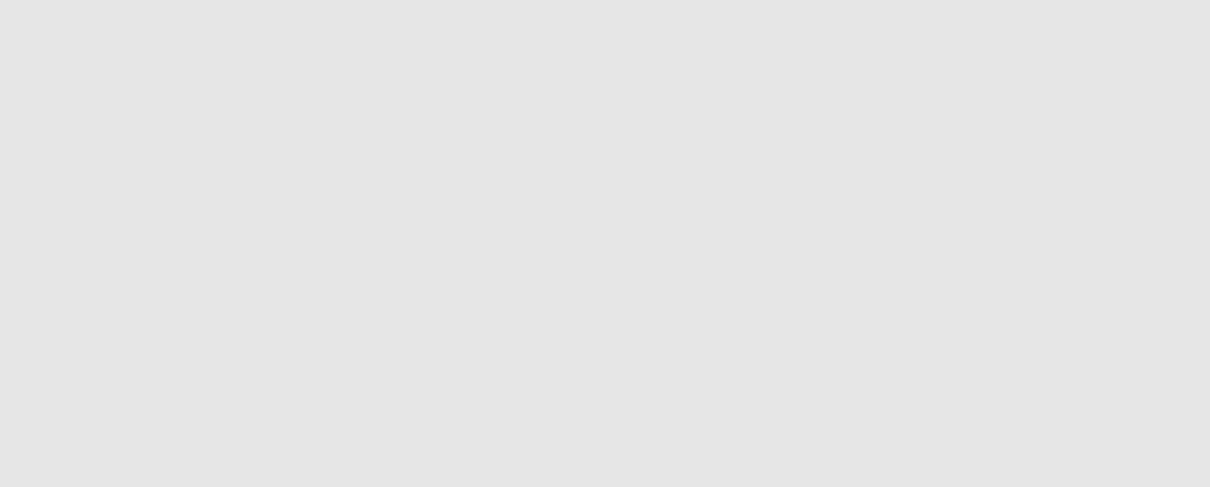
- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL**
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

2026-07-03

Prior-Data Fitted Networks
└ TabICL
 └ Outline

Outline

- Background
- TabPFN
- Statistical Foundations of PFNs
- TabPFNV2 & TabPFNV2.5
- TabICL**
- Recent SOTA Models
- Connections to Other Topics
- Conclusion



- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models**
- 7 Connections to Other Topics
- 8 Conclusion

2026-07-03

- Prior-Data Fitted Networks
 - Recent SOTA Models
 - Outline

Outline
Background
TabPFN
Statistical Foundations of PFNs
TabPFNV2 & TabPFNV2.5
TabICL
Recent SOTA Models
Connections to Other Topics
Conclusion

2026-07-03

Prior-Data Fitted Networks

└ Recent SOTA Models

└ Recent Sota Models

2026-07-03

Prior-Data Fitted Networks

└ Recent SOTA Models

└ Recent Sota Models

- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics**
- 8 Conclusion

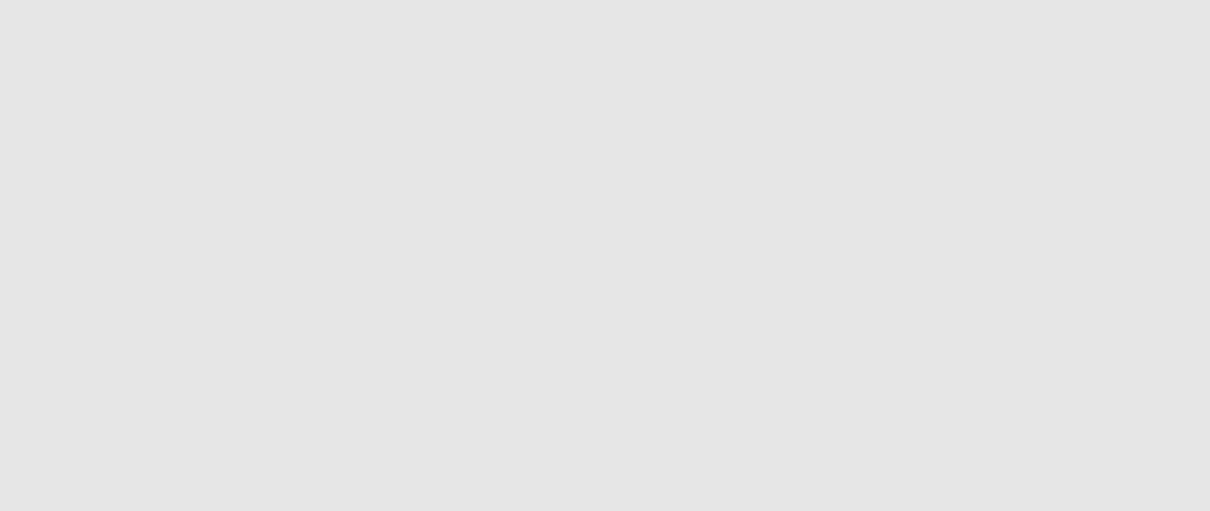
2026-07-03

Prior-Data Fitted Networks

- └─Connections to Other Topics
 - └─Outline

Outline

- Background
- TabPFN
- Statistical Foundations of PFNs
- TabPFNV2 & TabPFNV2.5
- TabICL
- Recent SOTA Models
- Connections to Other Topics**
- Conclusion



Connections to Other Topics

Background: Simulation-Based Inference

Connections to Other Topics

2026-07-03

Prior-Data Fitted Networks

—Connections to Other Topics

—Connections to Other Topics

2026-07-03

└─Connections to Other Topics

—Connections to Other Topics

2026-07-03

└─Connections to Other Topics

—Connections to Other Topics

- 1 Background
- 2 TabPFN
- 3 Statistical Foundations of PFNs
- 4 TabPFNV2 & TabPFNV2.5
- 5 TabICL
- 6 Recent SOTA Models
- 7 Connections to Other Topics
- 8 Conclusion

2026-07-03

Prior-Data Fitted Networks
└ Conclusion
└ Outline

- Background
- TabPFN
- Statistical Foundations of PFNs
- TabPFNV2 & TabPFNV2.5
- TabICL
- Recent SOTA Models
- Connections to Other Topics
- Conclusion

- Summary point one
- Summary point two

2026-07-03

Prior-Data Fitted Networks
└ Conclusion
└ Conclusion

- Conclusion
- Summary point one
 - Summary point two

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2026-07-03

Prior-Data Fitted Networks

└─ Conclusion

└─ References